Zoology Legends

ZOO405

Current Final TERM Spring 2023

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Zoology Legends

MCQS Spring 2023

- ✓ Autotrophs ("**self-feeders**"), can make their own food
- ✓ **Namib** desert is present in___**Africa**
- ✓ The First Law of Thermodynamics, also known as the **law of conservation of energy**
- ✓ **Ecology** is the study of organisms, the environment
- ✓ Abiotic components of an ecosystem include all chemical and physical elements i.e. **non-living** component Abiotic components
- ✓ Sublimation occurs when snow or ice changes directly Into **water vapors** without becoming water.
- ✓ **Trophic level** of an organism is the number of steps it is From the start of the chain.
- ✓ Marine ecosystem covers more than **70%** of the Earth's surface.
- ✓ Forests can be divided into **5** main categories
- ✓ **Ecosphere** (also sometimes called the **biosphere)**
- ✓ One third of earth surface is covered with **deserts**
- ✓ Which main factor decrease population? (**Death and emigration)**
- ✓ Abiotic component include (**hydro** + litho + atmosphere)
- ✓ ___ study of inland water and their many different aspects.(**Limnology**)
- ✓ **Coral reef** are the special type of marine ecosystem on seafloor.

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- ✓ The Earth is a system consisting of **four** major interacting components.
- ✓ The marine ecosystem covers more than **70% of** the Earth's surface.
- ✓ The inland waters include lakes, reservoirs, rivers, streams, and wetlands. The subject discipline, termed **limnology**.
- ✓ Producers are known as **autotrophs**
- ✓ **Abiotic components** are non-living chemical and physical factors of an ecosystem.
- ✓ fungi is **heterotroph**
- ✓ One third of earth consist of **desert**.
- ✓ Grassland occupy **20%** the land on the earth surface.
- ✓ The ecosphere also sometimes called the ' **biosphere**
- ✓ **Landscape** ecology deals with the exchange of energy, materials, organisms and other products of ecosystems.
- ✓ Ecologist name **Charles Elton**
- ✓ Abiotic factors could be acquired from all of these **atmosphere, lithosphere and hydrosphere**
- ✓ The hydrosphere includes all forms of water in the **Earth's environment**.
- ✓ IAEA stands for_____**International Atomic Energy Agency**
- ✓ **Desertification, also called desertization**, ____the process by Which natural or human causes reduce the biological Productivity of drylands

Zoology Legends

- ✓ **Exotic** species **harmless** while invasive species **Harmful** to the ecosystem,
- ✓ **Soil pollution** is a form of land pollution in which the Soil's __**upper layer is harmed.**
- ✓ Ozone absorb around __**97-99%**__ of the harmful ultraviolet radiation
- ✓ **Greenhouse Gases** ___absorb the energy radiated out by the Earth and Reflect a part of it back to Earth.
- ✓ **Competition** is a relationship between organisms in which one is harmed when both are trying to use the Same resource related to growth, reproduction, or Survivability.
- ✓ **Parasitism**__ is a symbiotic relationship in which one Species (the parasite) benefits while the other species (the host) is harmed.
- ✓ The most common form of weather modification is__ **Cloud seeding**
- ✓ **Aspects of applied ecology** include: Ecosystem restoration, Environmental engineering, Environmental technology, Habitat management___**all**
- ✓ Community's interactions include _& Competition, Predation, Herbivory, Symbiosis and disease. **All of these**
- ✓ Tundra ecosystem also known as the **Winter**.
- ✓ **Carbon** present in the atmosphere is absorbed by **plants** for photosynthesis.

Zoology Legends

- ✓ **Nimb** desert in Africa
- ✓ 1st law of thermodynamics also known as **law of conversation of energy**.
- ✓ **Wet land** act as nature kidney
- ✓ **Ecotone** act as a boundary or transition between two ecosystem.
- ✓ **Denitrification** ____is the process in which the nitrogen Compounds make their way back into the atmosphere Into gaseous nitrogen (N)
- ✓ In force since **2005**, the protocol called for reducing the **Emission of six greenhouse gases.**
- ✓ **Sulphur dioxide** is example of primary pollutant.
- ✓ **Pesticides** —chemical poisons used to kill weeds and Insects—seep into waterways and harm wildlife.
- ✓ **Cloud seeding** is a common technique to enhance precipitation.
- ✓ The first desalination technology to be developed was **thermal distillation**
- ✓ A **population** is the total number of individuals of a Given biological species found in one place at one time
- ✓ **Profundal Zone:** This zone extends beyond the **Limnetic** zone and is a Deeper region of the water. Here, there is usually very Little or no sunlight. It is cooler. Living things in this
- ✓ Region depend on drifting organic matter for food__**Profundal zone.**

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- ✓ **Noise pollution** is undesirable and unpleasant sound which leads to discomfort in human being.
- ✓ **Predation** is a biological interaction which one organism, predator kill and eats another organism it's prey.
- ✓ Lotic water includes **spring, stream river's**.
- ✓ Freshwater having low concentration usually less than **1% salt Concentration.**
- ✓ **Green House gasses** means to prevent **ultraviolet radiation**
- ✓ **Populations** _____ do not usually exist in isolation.
- ✓ There are_____ basic types of symbiosis. **3**
- ✓ A community's interactions include_____. **All options are correct (symbiosis, predator, and competition).**
- ✓ Different species living in same place may require the **Different** resources
- ✓ A _____ is an organism that consumes all or part of the body of another…**Predator**
- ✓ The denser a population is, the _____ its mortality rare…**Greater**
- ✓ Herbivory is the consumption of _____ by animals…**Plants**
- ✓ Density independent factors are ____. **All the option** are correct (drought, fire, flood).
- ✓ An_____ is on area that acts as a boundary or a transition between two ecosystems…**Ecotone**
- ✓ A community include all the ____ organisms in an area… **Living**

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- ✓ Sewage leakages, oil spillage and high population density ore the sources of **water pollution.**
- ✓ Any extensive area of land that is occupied by native herbaceous of shrubby vegetation which is grazed by domestic or wild herbivores is called ----------**Rangeland**
- ✓ **Spain**____Provides a significant example of the application of desalinated water in irrigation.
- ✓ Intensive research for large scale commercial desalinating technologies began in…**America**
- ✓ **silver iodide** _____ is used for freezing and crystals formation in cloud seeding process.
- ✓ The harmful materials for environmental pollution are called---------- **Pollutants**
- ✓ Applied ecologists often Use one or more of the approaches including----------**All** (modeling, experimentation, observation)
- ✓ According to United Nations, ____ million people do not have access to clean water…**783**
- ✓ **Applied ecology**_____ is a sub-field within ecology that considers the application of the science of ecology to real-world (usually management) questions
- ✓ Clouds were seeded to prolong the monsoon in___**Vietnam**
- ✓ IAEA stands for_____ **(International Atomic Energy Agency)**
- ✓ Desertification, also called **desertization,** ____the process by Which natural or human causes reduce the biological Productivity of dry lands

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- ✓ Exotic species harmless while invasive species **Harmful to the ecosystem**,
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- ✓ Aspects of applied ecology include: Ecosystem restoration, Environmental engineering, Environmental technology, Habitat management ___**all of these**
- ✓ Community's interactions include _&Competition, Predation, Herbivory, Symbiosis and disease. **All of these**
- ✓ Carbon present in the atmosphere is absorbed by plants for **photosynthesis.**
- ✓ Carrying capacity __**maximum** __number of organism exist.
- ✓ Lake not freeze having ____ **highest density.**
- ✓ **Tundra ecosystem** has long **cold winter.**
- ✓ Number of organisms per unit area is called___ **population density**
- ✓ Invasive species are ___ **harmful to ecosystem**.
- ✓ Which ecosystem act like kidney_____ **wetland**.
- ✓ Conversion of NH3 into gaseous is called as ____ **denitrification**
- ✓ Suppression of noise at source is way to control _____ **noise pollution.**
- ✓ **BOD** is rapid depletion of oxygen in ocean.

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- ✓ Percentage of phosphorus in teeth is ___**80%**
- ✓ **Applied ecology** is a sub field within ecology that considers the application of the science of ecology to real world questions.
- ✓ **Cloud seeding** is is a common technique to enhance ____ **precipitation.**
- ✓ **Pesticides** are chemical poisons used to kill weeds and insects seep into water ways and harm wildlife.
- ✓ Control at receiver's end is related to___ **noise pollution**
- ✓ SO2 is ____ **primary pollutant.**
- ✓ The denser a population the ___ **greater**___ its mortality rate.
- ✓ Grassland occupy about __ **20%** of land on earth surface.
- ✓ Agricultural ecosystem occupies __**30%**__ of earth surface.

ZOOLOGY
LEGENDS

Zoology Legends

Repeated Questions Spring 2023

Biotic and Abiotic Factors

• Biotic and abiotic factors include the Living and non-living factors and their Interaction with the environment.

Biotic components

Biotic components are living factors of an ecosystem.

• Biotic components are the living things that have a direct or indirect influence on other organisms in an environment. For example plants, Animals, and microorganisms and their waste Materials.

A few examples of Biotic components include bacteria, animals, birds, fungi, plants, etc.

Abiotic components

• Abiotic components are non-living chemical And physical factors of an ecosystem.

Abiotic components of an ecosystem include all chemical and physical elements i.e. non-living components. Abiotic components can vary from region to region, from One ecosystem to another.

They mainly take up the role of Life supporter. They determine and restrict the population growth, number,

And diversity of biotic factors in an ecosystem. Hence, they Are called limiting factors. These components could be acquired from the Atmosphere, lithosphere and hydrosphere.

A **Few examples** of abiotic components include Sunlight, soil, air, moisture minerals and more.

Importance of Ecology

Conservation of Environment

 Ecology helps us to understand how our actions affect the

Environment. It shows the individuals the extent of

Damage we cause to the environment.

 Lack of understanding of ecology has led to the

Degradation of land and the environment. It has also led to

The extinction and endangerment of certain species. For Eg., dinosaurs, white shark, mammoths, etc. Thus, the

Study of the environment and organisms helps us to

Protect them from any damage and danger.

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Producer's autotrophs, make their own organic Molecules.

• These include all the autotrophs. They use light energy And synthesize food on their own, e.g. plants, green Algae, etc.

• Producers are those organisms that use photosynthesis To capture energy by using sunlight, water and carbon Dioxide to create carbohydrates, and then use that Energy to create more complex molecules like proteins, Lipids and starches that are crucial to life processes.

Food chain	Food Web
Food chain is a linear sequence of organisms through which nutrients and energy pass as one organism eats another. In a **food chain**, each organism occupies a different trophic level, defined by how many energy transfers separate it from the basic input of the chain.	**Food webs** consist of many interconnected food chains and are more realistic representation of consumption relationships in ecosystems. Energy transfer between trophic levels is inefficient (with a typical efficiency around 10 %). This inefficiency limits the length of food chains.

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Autotrophs ("self-feeders"), can make their Own food – that is, their own organic Compounds.

Two basic types of autotrophs:

1. Photoautotrophs:

Such as plants, use energy from sunlight to make Organic compounds (sugars) out of carbon dioxide In photosynthesis.

 Other examples of photoautotrophs include algae And cyanobacteria.

2. Chemoautotrophs:

Chemoautotrophs use energy from chemicals to build Organic compounds out of carbon dioxide (or similar Molecules). This is called chemosynthesis.

 For instance, there are hydrogen sulfide-oxidizing Chemoautotrophic bacteria found in undersea vent Communities (where no light can reach).

Nitrogen Cycle:

Nitrogen Cycle is a biogeochemical process through Which nitrogen is converted into many forms, Consecutively passing from the atmosphere to the soil To organism and back into the atmosphere.

 It involves several processes such as nitrogen fixation, Nitrification, denitrification, decay and putrefaction.

Process of the Nitrogen Cycle consists of The following steps

- Nitrogen fixation
- Nitrification
- Assimilation
- Ammonification
- Denitrification

Nitrogen Fixation:

- It is the initial step of the nitrogen cycle. Here, Atmospheric nitrogen (N2) which is primarily available in An inert form, is converted into the usable form -ammonia (NH3).
- During the process of Nitrogen fixation, the inert form of Nitrogen gas is deposited into soils from the atmosphere And surface waters, mainly through precipitation.
- The entire process of Nitrogen fixation is completed by Symbiotic bacteria, which are known as Diazotrophs, Azotobacter and Rhizobium also have a major role in this process.

Nitrification:

- In this process, the ammonia is converted into nitrate by The presence of bacteria in the soil.

Nitrites are formed by The oxidation of ammonia with the help of Nitrosomonas Bacteria species. Later, the produced nitrites are converted Into nitrates by Nitrobacter. This conversion is very Important as ammonia gas is toxic for plants.

 The reaction involved in the process of Nitrification is as Follows:

$2NH_3 + 3O_2 \rightarrow 2NO_2^- + 2H^+ + 2H_2O$

$2NO_2^- + O_2 \rightarrow 2NO_3^-$

Assimilation:

 Primary producers – plants take in the nitrogen Compounds from the soil with the help of their Roots, which are available in the form of Ammonia, nitrite ions, nitrate ions or ammonium Ions and are used in the formation of the plant And animal proteins. This way, it enters the food Web when the primary consumers eat the plants.

Ammonification:

 When plants or animals die, the nitrogen present in The organic matter is released back into the soil. The decomposers, namely bacteria or fungi present In the soil, convert the organic matter back into Ammonium. This process of decomposition Produces ammonia, which is further used for other Biological processes.

Denitrification:

 Denitrification is the process in which the nitrogen Compounds make their way back into the atmosphere By converting nitrate (NO3-) into gaseous nitrogen (N).

This process of the nitrogen cycle is the final stage And occurs in the absence of oxygen. Denitrification is Carried out by the denitrifying bacterial species Clostridium and Pseudomonas, which will process Nitrate to gain oxygen and gives out free nitrogen gas as A byproduct.

Evaporation:

 The sun is the ultimate source of energy, and it powers Most of the evaporation that occurs on earth. Evaporation Generally happens when water molecules at the surface of Water bodies become excited and rise into the air.

These Molecules with the highest kinetic energy accumulate into Water vapour clouds. Evaporation usually takes place Below the boiling point of water. Another process called **Evapotranspiration** occurs when evaporation occurs Through the leaves of plants. This process contributes to a Large percentage of water in the atmosphere.

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Precipitation:

 Above 0 degrees centigrade, the vapours will Condense into water droplets. However, it cannot condense without dust or other impurities.

Hence, Water vapours attach itself on to the particle's Surface. When enough droplets merge, it falls out of The clouds and on to the ground below. This process Is called **precipitation (or rainfall).** In particularly Cold weather or extremely low air pressure, the water Droplets freeze and fall as snow or hail.

Decomposers:

An organism that gets energy by breaking down the Remains of dead organisms and the wastes of living Things. Another kind of heterotrophs.

Examples:

Most fungi and bacteria fall into this category. **Earthworms and insects, mites, snails, worm, mushrooms,** also They are often called nature's recyclers They return carbon and nitrogen to the air and Soil.

Grassland Ecosystem:

• Grasslands are areas dominated by grasses. They Occupy about **20%** of the land on the earth surface.

• Grasslands occur in both in tropical and temperate Regions where rainfall is not enough to support the Growth of trees.

• Grasslands are found in areas having well-defined **hot And dry**, warm and rainy seasons.

Functions of Grassland Ecosystem:

- They perform certain functions. These are energy flow through the food chain Nutrient cycling (biogeochemical cycles).ecological succession or ecosystem development Homeostasis (or cybernetic) or feedback control mechanisms.
- To increase the fertility of the soil and to regulate The productivity of the ecosystem.
- To reduce the leaching of minerals due to low Rainfall.

Desert ecosystem:

The desert ecosystem is the **driest** terrestrial ecosystem in The world, therefore, it has **less biodiversity**. The plants in this desert ecosystem are usually short-lived In nature. They grow, germinate, and perish in a very short Time.

In this type of ecosystem, **thorny shrubs** are scattered Around. The roots of these plants extend far below the Ground. Their roots are located at a depth of 1 meter from The surface. As a result, they can easily collect rainwater.

In many cases, their leaves are transformed into thorns to Reduce the cost of water through evaporation The desert ecosystem has less density of plants due to Low rainfall. These regions are characterized by **less Than 255 mm** of rainfall and high or low temperatures.

The evaporation rate of this ecosystem is very high. **Dates, cotton, millet**, etc. are cultivated in different Places depending on the availability of water. Like other ecosystems, desert ecosystems are made up Of some biotic and abiotic elements.

Characteristics of Population Ecology:

- All species occur in groups called populations. There are certain characteristics that all populations have, such as population density, spatial distribution, and growth rate. These characteristics are used to classify all populations of organisms, including bacteria, animals, and plants.
- Population density one characteristic of a population is its population density, which is the number of organisms per unit area.
- Another characteristic of a population is called dispersion—the pattern of spacing of a population within an area.

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Characteristics of a Population

- **Population** - individuals inhabiting the same area at the same time
- **Population Dynamics:** Population change due to
  - **Population Size** - number of individuals
  - **Population Density** - population size in a certain space at a given time
  - **Population Dispersion** - spatial pattern in habitat
  - **Age Structure** - proportion of individuals in each age group in population

1. **Population Density:**
   - The number of organisms per unit area
2. **Spatial Distribution:**
   - Dispersion: The pattern of spacing a population within an area
     - 3 main types of dispersion
       - Clumped
       - Uniform
       - Random
   - The primary cause of dispersion is resource availability

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Explain Water Pollution:

 Water pollution happens when chemicals or dangerous foreign substances are introduced to water, including chemicals, sewage, pesticides and fertilizers from agricultural runoff, or metals like lead or mercury.

 According to the Environmental Protection Agency (EPA), 44% of assessed stream miles, 64% of lakes and 30% of bay and estuarine areas are not clean enough for fishing and swimming.

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 Water pollution isn't just a problem for the United States. According to United Nations, 783 million people do not have access to clean water and around 2.5 billion do not have access to adequate sanitation. Adequate sanitation helps to keep sewage and other contaminants from entering the water supply

 According to National Oceanic and Atmospheric Administration (NOAA), 80% of pollution in marine environment comes from the land through sources like runoff.

 Water pollution can also severely affect marine life. For example, sewage causes pathogens to grow, while organic and inorganic compounds in water can change the composition of the precious resource.

 According to the EPA, low levels of dissolved oxygen in the water are also considered a pollutant. Dissolved is caused by the decomposition of organic materials, such as sewage introduced into the water.

SOURCES OF WATER POLLUTION

 Sewage leakages

 High population density

 oil spillage

 Menace of Nipa palm and water hyacinth

 Industrial waste dumped into our waters

 Pollution of ground water through drilling activities

 Flooding during rainy season which carries waste deposits into our waters.

 Water pollution is generally induced by humans. It results from actions of humans carried on to better self. These could be treated under the various activities that man engages in, that lead to pollution.

Managements and Control OF WATER POLLUTION:

 There are many approaches that could be adopted for management of water pollution and its control. It could be through prevention, practice efforts or join a project/program; Regulation and monitoring or engaging in control measures by reducing or minimizing waste.

 Wash your car far away from any storm water drains.

 Don't throw trash, chemicals or solvents into sewer drains

 Inspects your septic system every 3 – 5 years

- Avoid using pesticides and fertilizers that can run off into water systems
- Sweep your driveway instead of hosing it down
- Always pump your waste-holding tanks on your boat
- Use non-toxic cleaning materials

How to Prevent WATER POLLUTION?

- Turn off the tap when running water is not necessary
- Use environmentally household products
- Avoid over dosage of pesticides and fertilizers
- Don’t throw the non-degradable substances to rivers, lakes or oceans
- Pretreatment of industrial waste water before leaving to water bodies.

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Remember me in you Precious Prayer

Jazak Allah